

School Uniform & Supplies Marketplace - Enterprise Database Architecture

Core Modules

1. Vendor Management
2. Product Category Management
3. Product Management
4. Product Variant Management (Size, Color, SKU)
5. Product Image Management
6. School Approval & Product Mapping
7. Parent & Student Management
8. Orders & Order Items
9. Inventory & Stock Ledger
10. Warehouse Management
11. Audit Logs & Activity Tracking

Recommended Tables

vendors
categories
products
product_variants
product_images
schools
school_product_mappings
parents
students
orders
order_items
inventory_transactions
warehouses
warehouse_stock
audit_logs

Product Variant Structure

Every sellable item should be stored as a Product Variant.

Example:

Shirt -> Size S -> Stock 50

Shirt -> Size M -> Stock 80

Shirt -> Size L -> Stock 100

Each variant should contain SKU, Price, Discount Price, Stock Qty, Reorder Level.

Inventory Management

Do not store inventory directly in products table.

Use inventory_transactions table.

Types: Purchase, Sale, Return, Damage, Adjustment.

This provides complete stock history and auditability.

Order Flow

Parent -> Student -> School Approved Product -> Cart -> Order -> Payment -> Vendor Fulfillment.
Order Items table stores individual products purchased.

Scalability Recommendations

Use UUID primary keys.
Store images in separate table.
Add warehouse support from day one.
Maintain inventory ledger.
Use audit logs for every stock adjustment.
Support academic year versioning in school mappings.

Final Architecture Diagram

Vendor → Products → Product Variants → Inventory Transactions

Schools → School Product Mapping → Approved Catalog

Parents → Students → Orders → Order Items

Warehouses → Warehouse Stock → Inventory Ledger

Audit Logs track all operations.